

DATA ANALYSIS INTERNSHIP

Task 11

Video Game Industry Analysis

What Sells in the Video Game Market?
Genres, Platforms, Publishers, Regions and Ratings
of 16,715 Releases (1980–2016)

Data Quality • Cleaning • Exploratory Analysis • Statistical Tests • Power BI Dashboard •
Recommendations

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1 Executive summary

A company of the gaming industry wants to understand the video game market. This report analyses a dataset of game sales to answer seven questions: which genres are the most common, which platforms and publishers sell the most, which games are the best sellers, how sales have changed over time, which regions buy, and whether ratings go with sales. The notebook `Video-Game-Analytics.ipynb` reproduces every number and figure; the cleaned dataset is in `data/processed/`, and the Power BI dashboard in `powerbi/`.

Data. The main file lists **16,719 releases** (one game on one platform) on 31 platforms from 1980 to 2016, with sales in four regions, critic and user scores and the ESRB age rating (VGChartz sales, Metacritic scores, extraction of 22 December 2016). Two additional files give a later snapshot of the PS4 and Xbox One catalogues. Sales are **millions of units**, not dollars. After cleaning, **16,715 releases of 11,563 games** remain, with 99.97% of the units.

Key figures. **8,918 million units** sold. **Action** is both the most common and the best-selling genre, the **PlayStation 2** the best-selling platform and **Nintendo** the leading publisher. The typical release sells 170,000 copies; only 12.4% pass one million.

Main findings.

- **A hit-driven market:** 6.6% of the releases make half of all units.
- **Frequency is not performance:** Action is the most common genre, but Platform and Shooter games sell the most per release; Adventure is crowded with small sellers.
- **Nintendo** sells 20% of all units with 2.5 million per release; the ten largest publishers hold 70% of the market.
- **Retail sales peaked in 2008** (672 million units) and then fell by 60%, as digital and mobile games, which are not tracked, grew.
- **North America buys half of the units;** Europe has risen to the first rank for 2016 releases; Japan is a distinct market that prefers role-playing games.
- **Critic scores go with sales** (Spearman $\rho = 0.39$), **user scores much less** ($\rho = 0.15$); the median release scored 90+ sells 13 times more than one scored below 50.
- **Games sell for years:** the same PS4 and Xbox One games sold 29% more after December 2016.

Dashboard. The findings are delivered in an eight-page interactive Power BI dashboard (Section 8, Figure 13): an overview with the KPI cards required by the brief, one page per group of questions, an update on the PS4 and Xbox One generation, the insights and an auditable data-quality page.

2 Data understanding

2.1 The three files

Table 1: Supplied files.

File	Content	Rows
Video_Games_Sales_as_at_22_Dec_2016.csv	main file: one row per release; name, platform, year, genre, publisher, sales in NA, EU, JP, other regions and global, critic score and count, user score and count, developer, ESRB rating	16,719
PS4_GamesSales.csv	later VGChartz snapshot of PS4 games: game, year, genre, publisher, sales by region	1,034
XboxOne_GameSales.csv	same for Xbox One (plus a rank column)	613

Two notions must be kept apart: a **release** is a game on one platform (one row, 16,715 after cleaning) and a **game** is a title, all platforms together (11,563). *Grand Theft Auto V*, for example, is one game with five releases.

2.2 Columns, meaning and types

Table 2: Columns of the main file, with the type read by pandas and the missing values.

Column	Meaning	Read as	Correct type	Missing
Name	title of the game	text	text	2
Platform	console or system code (PS2, X360, Wii, PC...)	text	text	0
Year_of_Release	year of release on that platform	decimal	integer	269
Genre	12 genres	text	text	2
Publisher	company that published the release	text	text	54
NA/EU/JP/Other_Sales	units sold by region (millions)	decimal	decimal	0
Global_Sales	units sold worldwide (millions)	decimal	decimal	0
Critic_Score	Metacritic press score (0–100)	decimal	decimal	8,582
Critic_Count	number of critics	decimal	integer	8,582
User_Score	Metacritic player score (0–10)	decimal	decimal	9,129
User_Count	number of players	decimal	integer	9,129
Developer	studio	text	text	6,623
Rating	ESRB age rating	text	text	6,769

Sales are extremely skewed: median 0.17, mean 0.53 and maximum 82.5 million units (*Wii Sports*). Critic and user scores exist for less than half of the releases, almost none before 1996: this is a coverage limit of Metacritic, not an error to correct.

3 Data quality and cleaning

3.1 Issues found

Table 3: Data quality audit and treatment.

Issue	Rows	Treatment
No name and no genre (Genesis, 1993)	2	removed (2.42 M units)
Split record of one release: <i>Madden NFL 13</i> (PS3), <i>Sonic the Hedgehog</i> (PS3)	2	merged (sales added)
Two different games with one name: <i>Need for Speed: Most Wanted</i> 2005 and 2012	2 pairs	kept, 2012 releases renamed
Year after the extraction date (2017, 2020)	4	year set to missing
Missing year	269	kept, <i>Unknown</i> , out of time series
Missing publisher	54	<i>Unknown</i>
Legacy ESRB code K-A (= E before 1998), pending code RP	3 + 3	K-A → E; RP and blank → <i>Not rated</i>
Years and counts stored as decimals	3 columns	integers
Exact duplicates; negative or zero sales	0	none found
Console files: Mac Roman encoding, carriage-return line breaks, non-breaking spaces, 512 zero-sales rows (announced games)	1,647	decoded, trimmed, zero-sales rows removed

The duplicate check shows why a mechanical rule is not enough: three name-platform pairs appear twice, but two of them are fragments of one release (same critic score, tiny or partial sales, missing year) and must be merged, while the third is a 2012 reboot of a 2005 game and must be kept. The 2020 release of *Imagine: Makeup Artist* on DS, a console discontinued in 2013, is a typing error: its year is set to missing rather than guessed. The console files, read with default settings, appear empty, because they use the old Mac line breaks; they are decoded with the Mac Roman encoding (byte 142 = é).

3.2 Cleaned dataset

Every step is recorded in a cleaning log (`data/processed/cleaning_log.csv`), also shown in the dashboard: **16,719 rows** → **16,715 releases**, units 8,920.30 → **8,917.88 million (99.97%)**. The cleaned dataset `video_games_clean.csv` has 25 columns: the original ones, plus platform name, manufacturer (Sony, Nintendo, Microsoft, Sega, PC, Other) and type (home console, handheld, PC), decade, user score $\times 10$ (same scale as the critic score), critic score band and sales tier. Three companion tables feed the dashboard: the regional sales in long format (66,860 rows), the cleaned console files (1,135 releases with sales) and the cleaning log.

4 Key performance indicators

Table 4: KPIs of the dashboard (no filter).

KPI	Value	Comment
Total games	11,563	distinct titles
Total releases	16,715	game × platform
Total sales	8,917.9 M units	1980–2016
Top genre	Action	also the most common (3,370 releases)
Top platform	PlayStation 2	1,256 M units
Top publisher	Nintendo	1,789 M units
Average / median sales per release	0.53 / 0.17 M	skewed by blockbusters
Releases above 1 M units	12.4%	

5 Exploratory analysis

5.1 Which genres are the most common? (Q1)

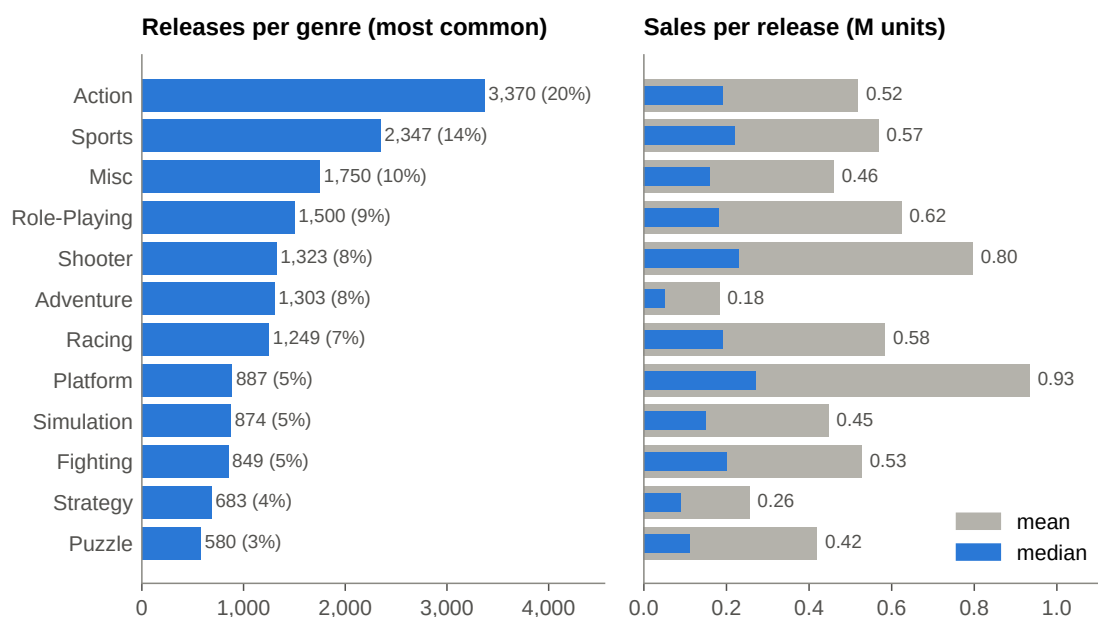


Figure 1: Releases per genre (left) and units per release (right; grey: mean, blue: median).

Action is the most common genre (3,370 releases, 20%), ahead of **Sports** (14%) and **Misc** (10%). Frequency and performance differ: **Platform** games are only the 8th most common genre but sell the most per release (mean 0.93 M, median 0.27 M), followed by **Shooter** (0.80 M). **Adventure** is the 6th most common genre but the weakest seller (median 0.05 M); **Strategy** and **Puzzle** are similar. In every genre the median is three to four times below the mean: each genre lives on a few hits.

5.2 Which platforms have the highest sales? (Q2)

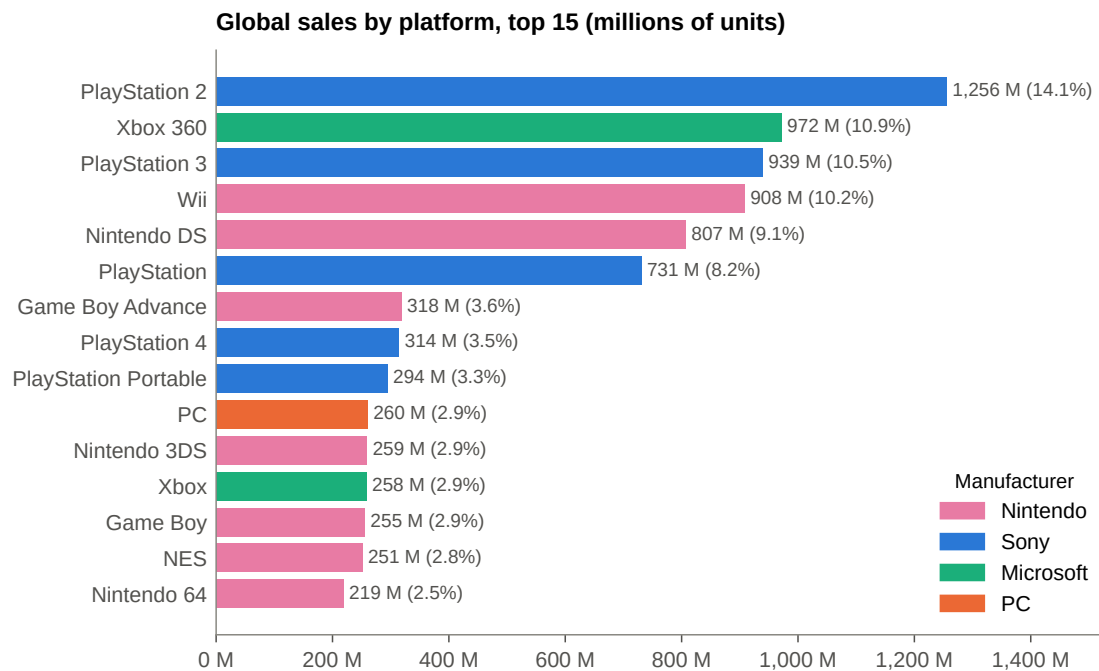


Figure 2: Global sales of the 15 best-selling platforms, coloured by manufacturer.

The **PlayStation 2** leads (1,256 M units, 14.1%), ahead of the Xbox 360 (972 M), PS3 (939 M), Wii (908 M) and DS (807 M), all consoles of the 2000s, the peak of retail sales. By manufacturer, **Sony (40%)** and **Nintendo (39%)** share four fifths of the market, Microsoft 16%. The PS4, launched in 2013, already ranks 8th: it leads the most recent period.

5.3 Which publishers have the highest total sales? (Q3)

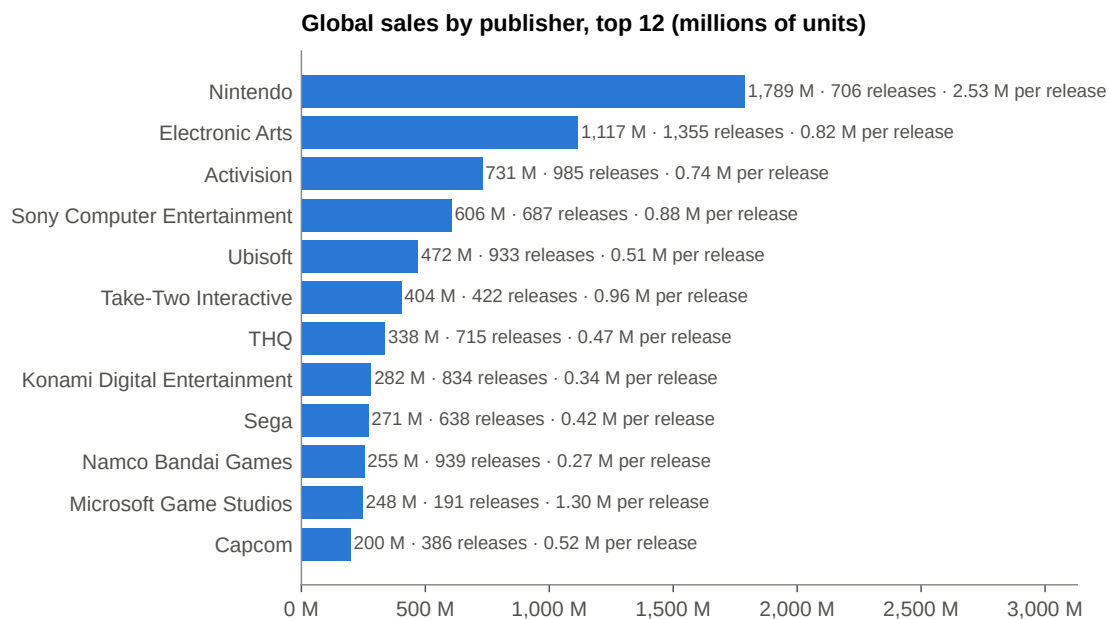


Figure 3: Global sales of the 12 largest publishers, with their number of releases and units per release.

Nintendo leads with 1,789 M units (20%), followed by Electronic Arts (1,117 M), Activision (731 M), Sony Computer Entertainment (606 M) and Ubisoft (472 M). Two models appear: Nintendo publishes few releases (706) that sell 2.53 M each, five times the average, on its own consoles; EA publishes almost twice as many (1,355) at 0.82 M each, with annual franchises on every platform. The market is concentrated: the 10 largest of 581 publishers hold **70%** of the units.

5.4 What are the best-selling games? (Q4)

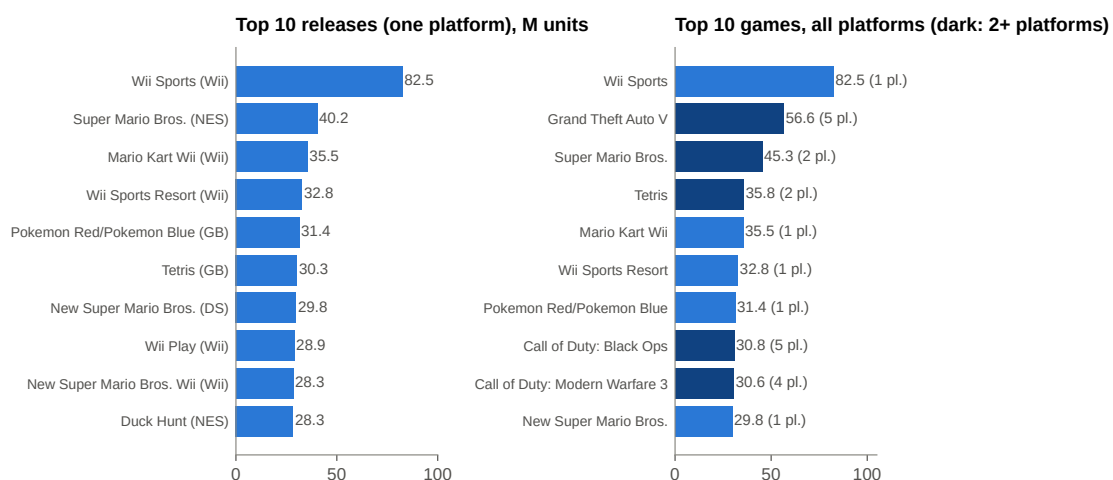


Figure 4: Ten best-selling releases (left) and ten best-selling games, all platforms together (right).

Wii Sports (82.5 M) is by far the best-selling release, and the ten best-selling releases are all Nintendo games on Nintendo hardware, several of them bundled with the console. Summing the platforms of each title changes the ranking: **Grand Theft Auto V** becomes the second best-selling game (56.6 M on 5 platforms), and two *Call of Duty* episodes (about 31 M each) enter the top 10. For a third-party publisher, the route to the top is the multi-platform blockbuster.

5.5 How have game sales changed over the years? (Q5)

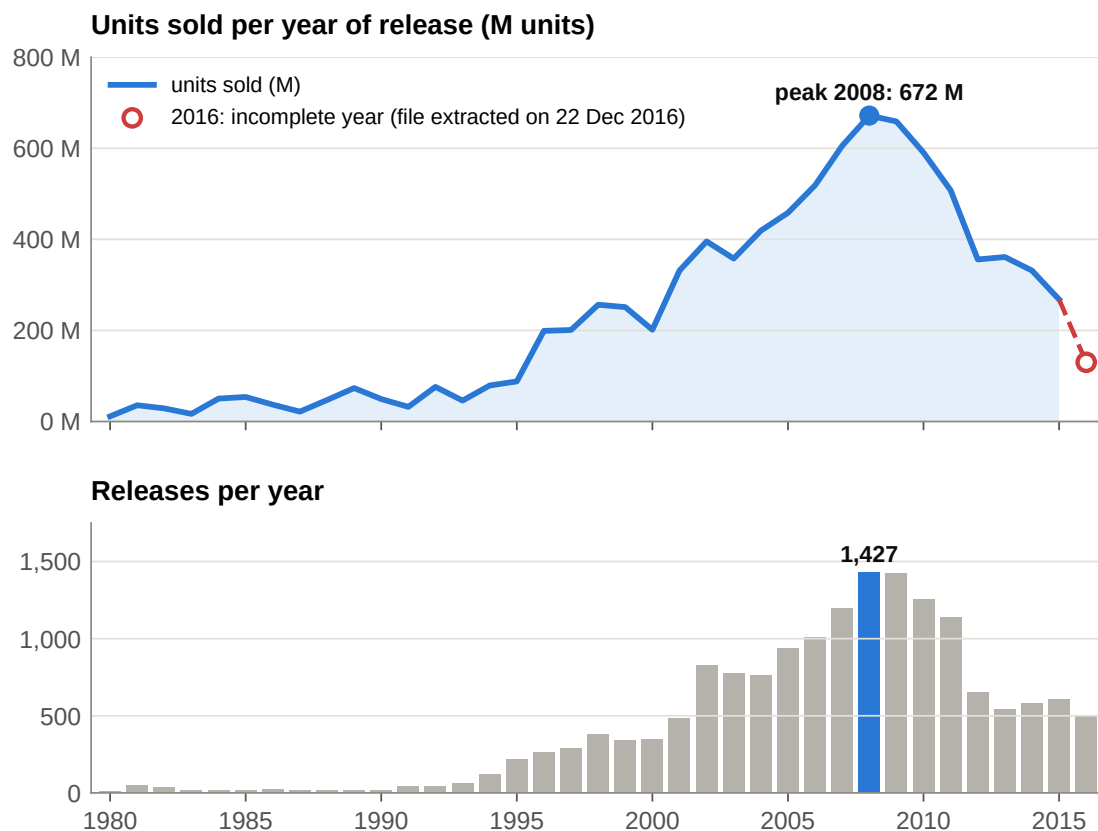


Figure 5: Units sold (top) and releases (bottom) per year of release.

The market went through three phases: a small market until 1995 (under 100 M units a year); fast growth from 1996 to the **peak of 2008** (672 M units, 1,427 releases); then a decline of tracked retail sales, to 268 M units in 2015 (−60%). Releases and units fell at the same pace, so each release sells about as much as before. The decline concerns *retail units*: digital downloads, mobile and free-to-play games are not tracked, and 2016 is incomplete.

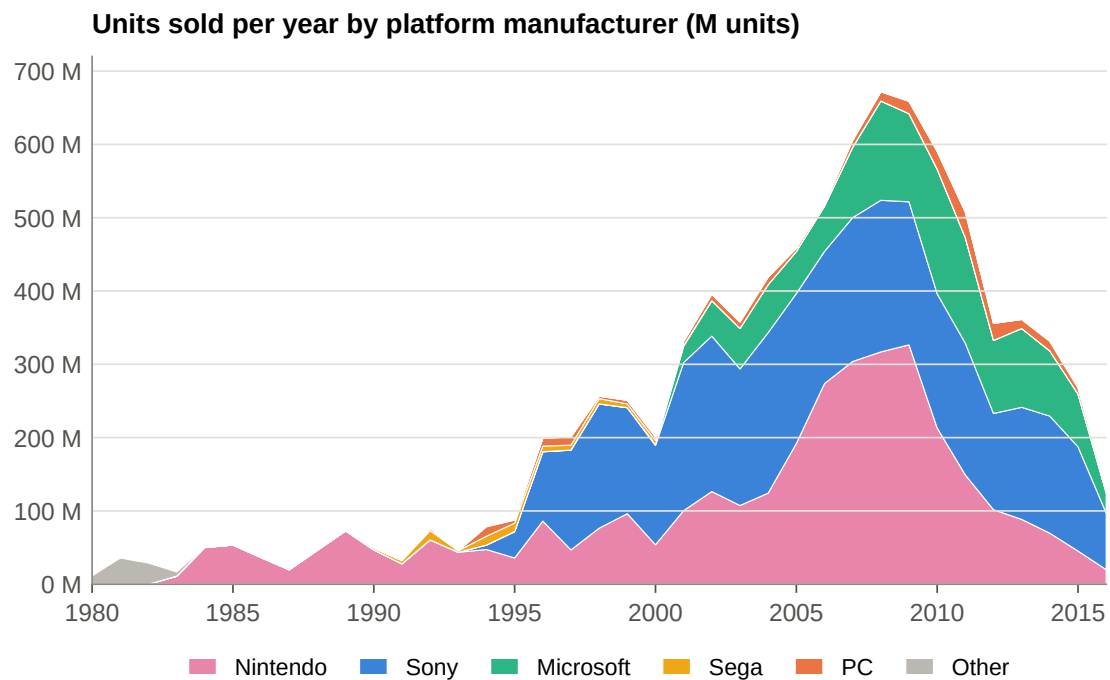


Figure 6: Units sold per year by platform manufacturer.

Sales follow **console generations** of six to eight years: Sony leads with the PlayStation and PS2, Nintendo peaks with the Wii and DS (2006–2010), Microsoft grows with the Xbox 360; Sega leaves hardware after 2001.

5.6 Which regions generate the highest sales? (Q6)

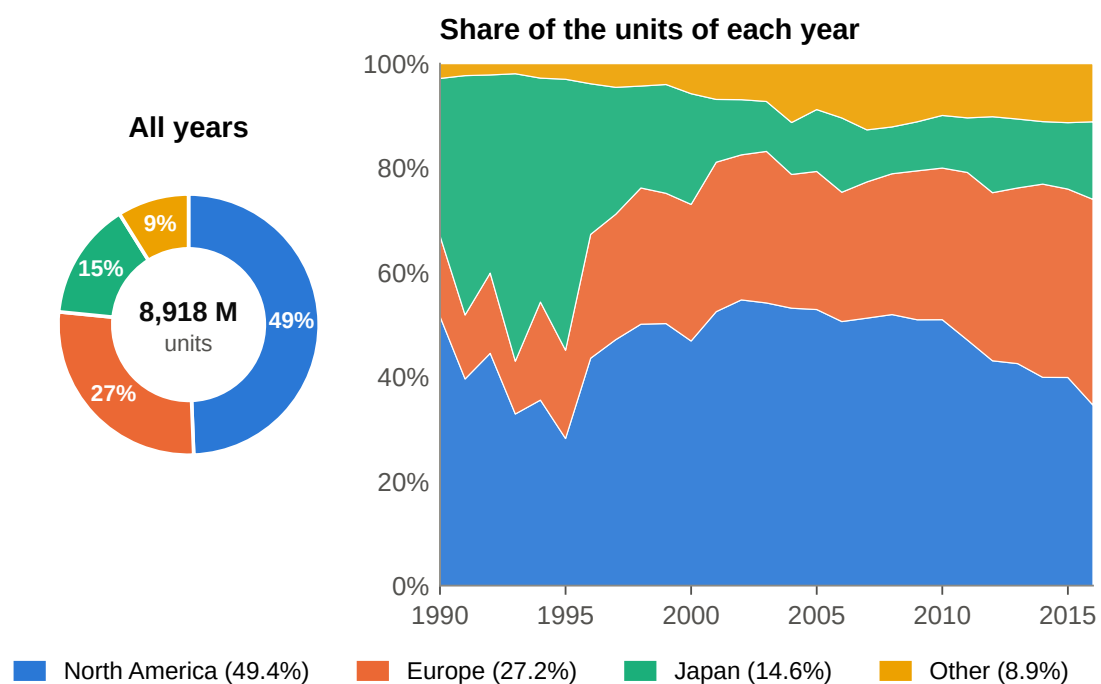


Figure 7: Share of units by region: all years (left) and per year (right).

North America generates half of all units (49.4%), ahead of Europe (27.2%), Japan (14.6%) and the rest of the world (8.9%). Japan weighed 27–29% in the 1980s and 1990s but 10–15% since 2005; Europe rose from 8% (1980s) to 33% (2010s) and is the first region for 2016 releases (39% vs 35%).

Share of each region's units by genre (each column adds up to 100 %)

	North America	Europe	Japan	Other
Action	20.0 %	21.4 %	12.4 %	23.3 %
Sports	15.6 %	15.5 %	10.4 %	17.0 %
Shooter	13.5 %	13.1 %	3.0 %	13.2 %
Platform	10.1 %	8.3 %	10.1 %	6.5 %
Misc	9.3 %	8.8 %	8.3 %	9.4 %
Racing	8.2 %	9.8 %	4.4 %	9.6 %
Role-Playing	7.5 %	7.8 %	27.4 %	7.5 %
Fighting	5.1 %	4.1 %	6.7 %	4.6 %
Simulation	4.1 %	4.7 %	4.9 %	3.9 %
Puzzle	2.8 %	2.1 %	4.4 %	1.6 %
Adventure	2.4 %	2.6 %	4.0 %	2.1 %
Strategy	1.6 %	1.9 %	3.8 %	1.4 %

red frame: where Japan differs most from the other regions

Figure 8: Share of each region's units by genre (each column adds up to 100%); the red frames mark the two cells where Japan differs most.

Japan is a different market: role-playing games make 27.4% of its units (7.5% in North America), shooters only 3.0% (13.5%). North America, Europe and the rest of the world buy the same mix of Action, Sports and Shooter games.

5.7 Is there a relationship between ratings and sales? (Q7)

The analysis uses the 8,135 releases with a critic score (48.7% of the releases but 62.9% of the units: reviewed games are the more visible ones).

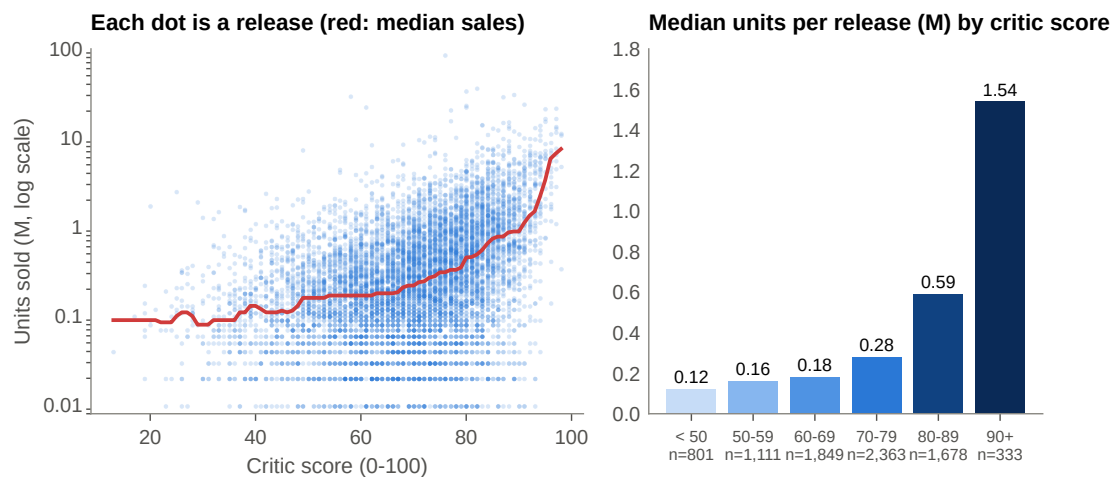


Figure 9: Critic score and units sold: every release (left, log scale) and median per score band (right).

Better-reviewed games sell more, and the effect accelerates at the top: the median release sells 0.12 M units below a score of 50, 0.28 M at 70–79, 0.59 M at 80–89 and **1.54 M at 90 or more**, thirteen times more than the lowest band. The cloud is wide, however: the score shifts the odds, it does not guarantee a hit.

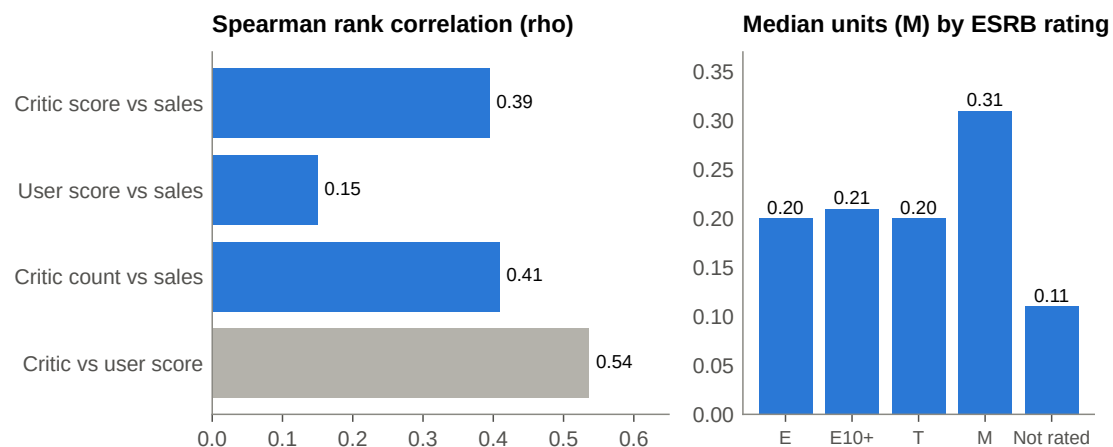


Figure 10: Rank correlations with sales (left) and median units by ESRB rating (right).

The press matters more than the players: $\rho = 0.39$ for the critic score, 0.15 for the user score, although the two scores agree ($\rho = 0.54$). Critic reviews appear at launch, when most copies are sold; user scores accumulate later. Mature (M) games have the highest median (0.31 M), unrated releases the lowest (0.11 M).

5.8 A hit-driven market

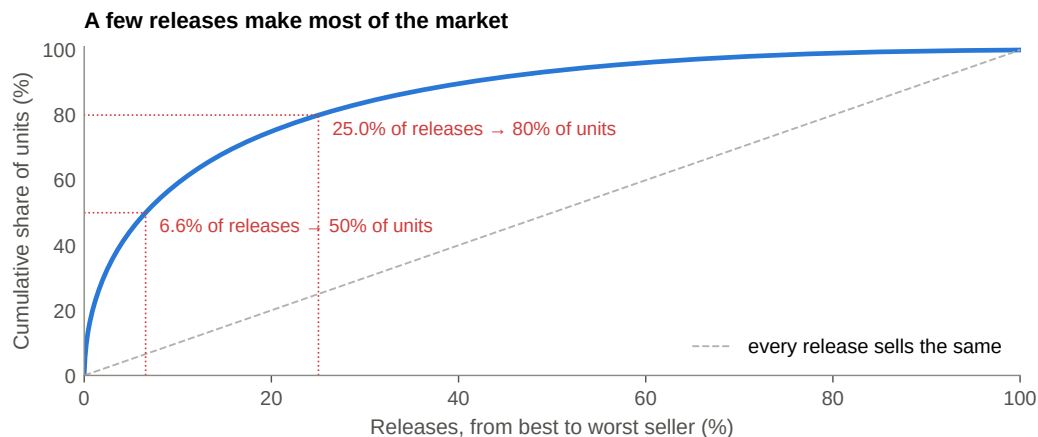


Figure 11: Cumulative share of units, releases sorted from best to worst seller.

6.6% of the releases make half of all units and one quarter makes 80%, while 35% of the releases sell fewer than 100,000 copies.

6 Statistical tests

Sales are very skewed, so the tests are rank-based. With thousands of releases even weak effects are significant; the size of the effect is what matters.

Table 5: Spearman correlations with global sales (95% bootstrap intervals) and Kruskal-Wallis tests.

Relationship	n	Effect	Reading
Critic score – sales	8,135	$\rho = 0.39$ [0.37, 0.41]	moderate, strongest
Critic score – sales, controlling for the number of critics	8,135	partial $\rho = 0.26$	survives the visibility effect
Critic score – sales within each genre	12 genres	ρ 0.13 to 0.50	positive in all genres
Number of critics – sales	8,135	$\rho = 0.41$	big games get more reviews
User score – sales	7,588	$\rho = 0.15$ [0.13, 0.17]	weak
Year of release – sales	16,443	$\rho = -0.16$	recent releases sell slightly less each
Sales by critic band (Kruskal-Wallis)	8,135	$\varepsilon^2 = 0.16$	large
Sales by genre / ESRB / manufacturer		ε^2 0.06 / 0.04 / 0.03	small

The critic score is the best single predictor of sales in this file, and the link holds once the number of critics (a proxy of visibility) is removed. These are associations, not causes: a large budget can produce both a better game and a larger marketing campaign.

7 After 2016: the PS4 and Xbox One generation

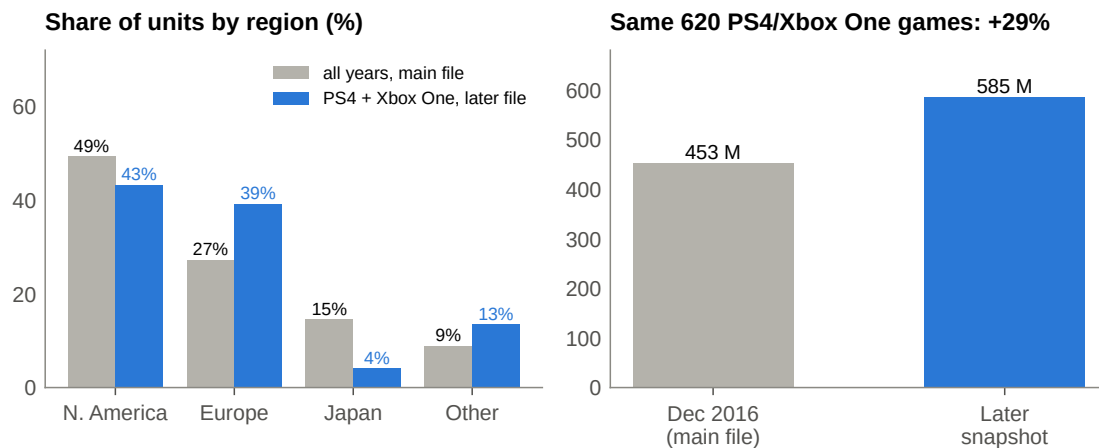


Figure 12: Region shares (left) and sales of the same 620 PS4 and Xbox One games in December 2016 and in the later snapshot (right).

The console files are not merged with the main file (other snapshot, other genre names); they answer two questions. **In the current generation, Japan almost disappears** (4% of PS4 and Xbox One units) and **Europe (39%) catches up with North America (43%)**; the leaders are Western blockbusters (GTA V, Call of Duty, Red Dead Redemption 2, FIFA). **Games keep selling**: the 620 releases found in both files sold 453 M units by December 2016 and 585 M in the later snapshot, **+29%** (median +22% per game); GTA V on PS4 went from 12.6 to 19.4 M. The 2015–2016 figures of the main file are therefore under-estimated, and the back catalogue has real value.

8 The Power BI dashboard

The dashboard is delivered as the template `powerbi/Video_Game_Analytics_Dashboard.pbix`. On opening, Power BI asks for DataFolder, the folder `data/processed/`, and loads the four cleaned tables, so every number matches the notebook. The model has five tables (Games, Regional Sales, PS4 XOne, PS4 XOne Regions, Cleaning Log), one relationship (Regional Sales → Games on Game ID) and 22 DAX measures. *Top Genre*, *Top Platform* and *Top Publisher* are computed with TOPN and follow the filters.

Table 6: Pages of the dashboard.

Page	Visuals
1. Overview	KPI cards (Total Games, Total Releases, Total Sales, Top Genre, Top Platform, Top Publisher); sales trend; region donut; sales by genre, top 10 platforms and publishers; scrolling best sellers
2. Genres & Platforms	releases by genre; average and median units per release by genre; units per year by manufacturer; top 12 platforms
3. Publishers & Games	top 10 publishers; releases vs units per release by publisher; top 10 releases; top 10 games, all platforms
4. Regions	region donut; region share by year; genre share inside each region; units by region
5. Ratings & Sales	score cards; median units by critic band; units at each critic score; critics vs players; ESRB rating
6. PS4 & Xbox One	cards of the console files; region donut; top 10 games; genres; units by year and console
7. Insights	the insights and recommendations of Section 9
8. Data Quality	pipeline summary and cleaning log

Design. The report uses a *Neon Arcade* theme made for the gaming audience: an illustrated night background with a synthwave grid, a header banner with the logo and a pill-shaped page navigator, KPI cards with icons, translucent rounded panels behind every visual and one neon colour per chart (cyan, magenta, violet, amber, lime). The palette is kept consistent: the four regions always have the same colours.

Interactivity. Four slicers synchronised across the analysis pages (year range, genre, manufacturer, platform); cross-filtering between visuals (a click on a bar filters the page); Top N filters that follow the slicers; the page navigator; tooltips on every visual. The build guide (`powerbi/Dashboard_Build_Guide.md`) lists the values to check after opening; every capture below, taken with no filter, shows these values.

8.1 Overview



Figure 13: Overview. The six KPI cards answer the brief at a glance: 11,563 games, 16,715 releases, 8,917.9 million units, and the three leaders (Action, PlayStation 2, Nintendo). The area chart shows the rise to the 2008 peak and the decline; the donut gives the regional split (North America 49.4%); the three bar charts rank genres, platforms and publishers; the ticker at the bottom scrolls the best-selling games (Wii Sports 82.5 M, GTA V 56.6 M).

The Overview is the one-page summary required by the brief. Every card is a DAX measure that follows the slicers: selecting the genre *Role-Playing*, for example, makes Nintendo DS the top platform, and selecting the manufacturer *Nintendo* makes the Wii the top platform.

8.2 Genres and platforms (Q1, Q2, Q5)

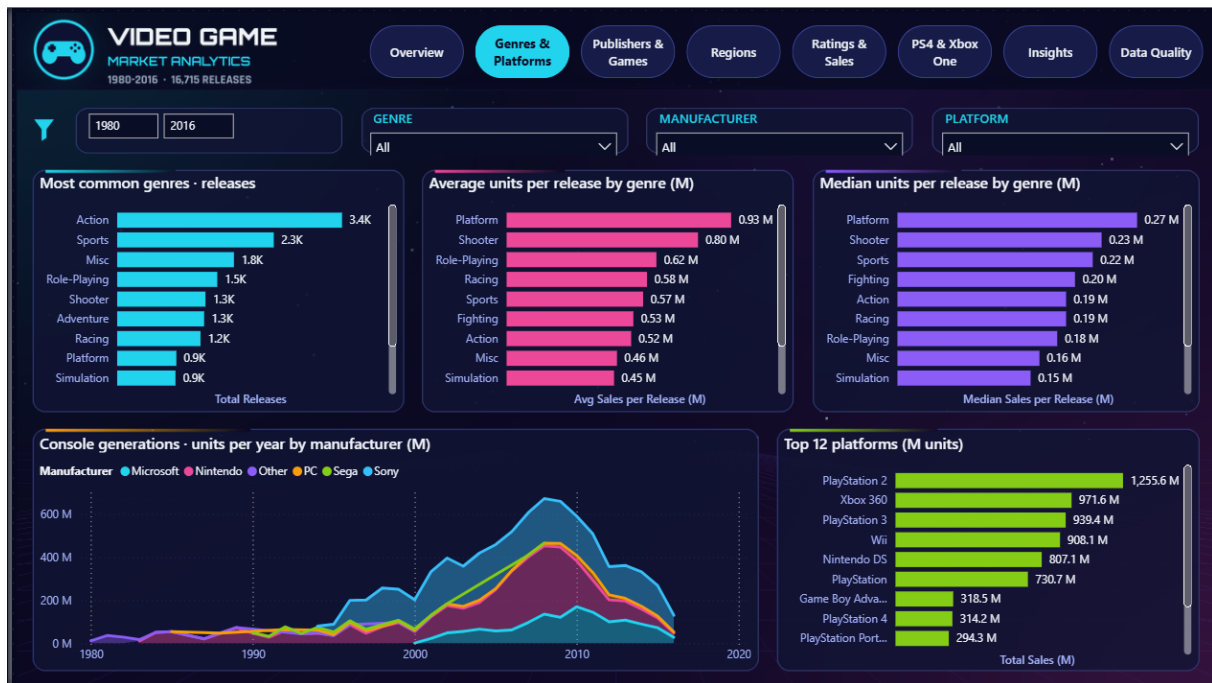


Figure 14: Genres & Platforms. Left to right: the most common genres (Action 3.4 K releases), the average and the median units per release by genre (Platform first in both: 0.93 M and 0.27 M), the stacked units per year by manufacturer (Sony and Nintendo dominate the 2000s peak) and the twelve best-selling platforms (PS2 1,255.6 M).

Putting frequency next to the average and the median makes the key nuance of Q1 visible: Action is the most common genre, but Platform and Shooter games sell the most per release, and the medians are three to four times below the means.

8.3 Publishers and games (Q3, Q4)



Figure 15: Publishers & Games. The top 10 publishers (Nintendo 1,788.8 M, EA 1,117.0 M); the scatter of releases against units per release separates the *hit* model (Nintendo, top left: few releases, 2.5 M each) from the *volume* model (Electronic Arts, right: 1,355 releases); the table of the best-selling releases and the top 10 games summed over platforms (GTA V second with 56.6 M).

8.4 Regions (Q6)



Figure 16: Regions. Share of units by region (North America 49.37%, Europe 27.2%, Japan 14.56%, Other 8.88%); share of each year's units (Japan's weight collapses after 1995, Europe grows to the first rank in 2016); genre share inside each region (Japan's role-playing peak and shooter gap) and units by region.

8.5 Ratings and sales (Q7)



Figure 17: Ratings & Sales. Cards: 8,135 scored releases, average critic score 69.0, average user score 71.3 (x10), median 0.24 M units. The median per critic band rises from 0.12 M (below 50) to 1.54 M (90+); the average units at each score bend upwards above 80; critics and players agree only loosely (wide cloud around the trend line); by ESRB rating, Mature games lead among the large groups.

Two readings need care on this page. The *No score* band (0.12 M) gathers the releases never reviewed by Metacritic, mostly older or smaller games. In the ESRB chart, **AO (1.95 M)** is a single release and **EC** has only eight: the meaningful comparison is between **M (0.31 M)**, **E10+**, **E** and **T** (about 0.20 M) and *Not rated* (0.11 M).

8.6 After 2016: PS4 and Xbox One



Figure 18: PS4 & Xbox One (later VGChartz snapshot). Cards: 1,135 releases with sales, 864.7 M units, 0.76 M per release, Japan 4.1% of units. The donut shows North America 43.3% and Europe 39.1%; GTA V leads with 28.1 M units on the two consoles; shooters lead the genres (227.2 M); the PS4 outsells the Xbox One every year.

8.7 Insights and data quality

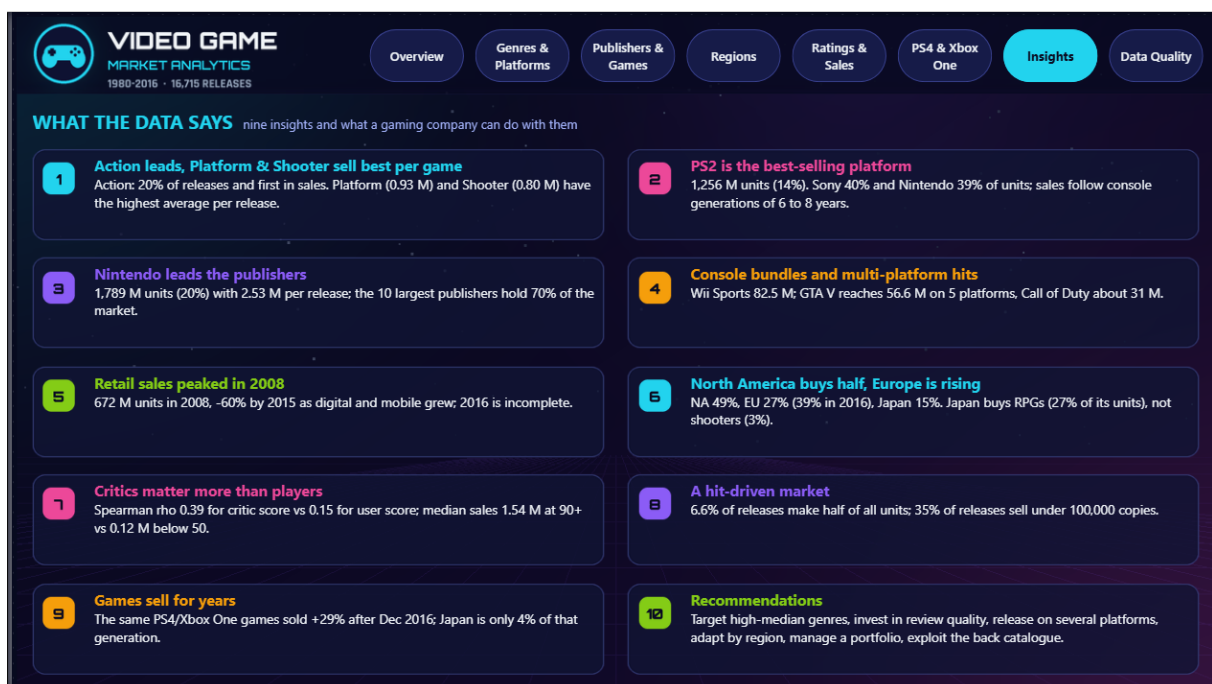


Figure 19: Insights: the nine findings of the analysis and the recommendations, each tile with its key figures.

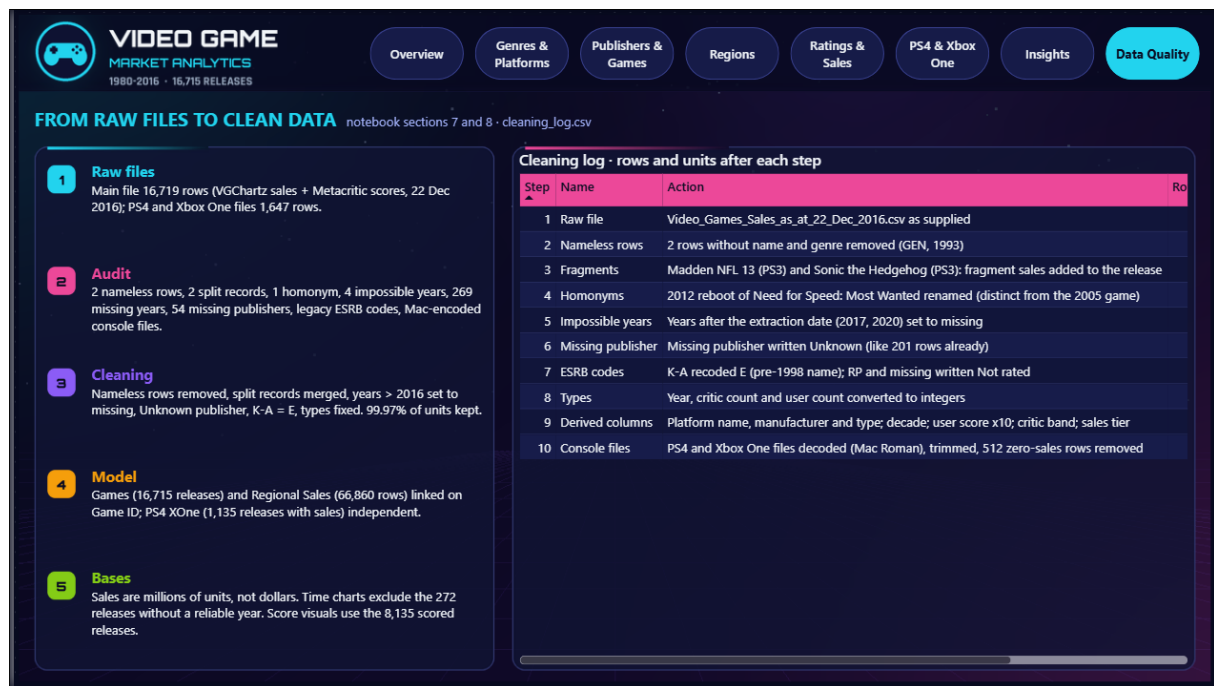


Figure 20: Data Quality: the five steps from the raw files to the model, and the cleaning log loaded from `cleaning_log.csv` (16,719 rows to 16,715 releases, 99.97% of units kept).

The last page makes the cleaning auditable inside the dashboard itself: a reader can see which rows were removed, merged or recoded, and on which base each visual is computed.

9 Insights and recommendations

1. **Choose genres by return per release, not by popularity.** Platform, Shooter, Sports and Racing have the highest medians; crowded genres with low medians (Adventure, Strategy, Puzzle) need a clear differentiation or a small budget.
2. **Invest in quality as judged by the press.** Median sales jump above a critic score of 80 (0.59 M) and 90 (1.54 M): polish, review-ready launches and press relations pay; user scores are a weaker signal.
3. **Release on several platforms.** GTA V and Call of Duty reach the top by adding platforms and generations; plan ports early for the leading platforms of the current generation.
4. **Adapt the offer to each region.** Shooters and sports for North America and Europe; role-playing games, handheld and mobile formats for Japan; give Europe the same launch effort as North America.
5. **Manage a portfolio of bets.** Half of the market is made by 6.6% of releases: plan with medians and hit probabilities, fund a few potential blockbusters, keep small titles cheap.
6. **Exploit the back catalogue:** established games still sell 29% more within about two years (price cuts, bundles, remasters, ports).
7. **Complete the data with digital sales** before deciding on the future of a genre or platform:

retail units no longer describe the whole market.

10 Limitations

- Retail units only: no prices, revenue, digital, mobile or free-to-play sales.
- VGChartz figures are estimates and are revised; 2016 is incomplete.
- Scores exist for half of the releases, almost none before 1996.
- The correlations are associations, not causes.

11 Reproducibility

Every number, table and figure of this report is produced by the notebook `Video-Game-Analytics.ipynb` from the three files of `data/raw/`. The notebook writes the cleaned dataset and the dashboard tables (`data/processed/`), the result tables (`results/`) and the figures (`figures/`). The template `powerbi/Video_Game_Analytics_Dashboard.pbix` loads `data/processed/`. The project is published at <https://github.com/koffifidelek59-collab/video-game-analytics>.

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